第 4 章总复习题(1-10) 参考答案

1. (2) $\exists x_n \in (2^n, 2^{n+1}), s.t. \dfrac{f(2^{n+1}) - f(2^n)}{2^{n+1} - 2^n} = f'(x_n).$

2. 又且仅有 1 个零点.

存在性:

$f(0) = 0, f'(0) < 0, 则存在 x_0 > 0, s.t. f(x_0) < 0.$

$f(x) = f(0) + f'(0)x + \dfrac{f''(\xi_x)}{2}x^2 \geq f'(0)x + \dfrac{a}{2}x^2, 则 \lim\limits_{x \to +\infty} f(x) = +\infty.$

唯一性: 反证法.

3. 由高阶导的 Leibniz 公式, n 阶 Laguerre 多项式

$$L_n(x) = e^x \frac{\mathrm{d}^n}{\mathrm{d}x^n}(x^n e^{-x}) = e^x \sum_{k=0}^{n} C_n^k (e^{-x})^{(k)} (x^n)^{(n-k)} = \sum_{k=0}^{n} C_n^k (-1)^k (x^n)^{(n-k)}$$

是首项系数为 $(-1)^n$ 的 n 次多项式, 至多有 n 个根, 且 $L_n(0) = n! > 0.$ 下面归纳证明 $L_n(x)$ 恰有 n 个相异正根.

$n = 1$时, $L_1(x) = -x + 1,$ 恰有一个正根 $x = 1.$

$n = 2$时, $L_2(x) = x^2 - 4x + 2,$ 恰有两个相异正根 $x = 2 \pm \sqrt{2}.$

设 $n = m$时, $L_m(x) = (-1)^m \prod\limits_{k=1}^{m}(x - x_k)$ 恰有 m 个相异正根 $(0 <) x_1 < x_2 < \cdots < x_m,$ 即

$$(x^m e^{-x})^{(m)} = (-1)^m e^{-x} \prod_{k=1}^{m}(x - x_k).$$

则 $n = m + 1$时,

$$\begin{aligned} e^{-x} L_{m+1}(x) &= (x \cdot x^m e^{-x})^{(m+1)} = x \cdot (x^m e^{-x})^{(m+1)} + (m+1)(x^m e^{-x})^{(m)} \\ &= x \cdot \left((-1)^m e^{-x} \prod_{k=1}^{m}(x - x_k) \right)' + (m+1)(-1)^m e^{-x} \prod_{k=1}^{m}(x - x_k). \end{aligned}$$

$$(-1)^m e^{-x} L_{m+1}(x) - (m+1) e^{-x} \prod_{k=1}^{m}(x - x_k)$$

$$= x \cdot \left(e^{-x} \prod_{k=1}^{m}(x - x_k) \right)' = x e^{-x} \left(\left(\prod_{k=1}^{m}(x - x_k) \right)' - \prod_{k=1}^{m}(x - x_k) \right)$$

$$= x e^{-x} \left((x - x_i) \left(\prod_{\substack{k=1 \\ k \neq i}}^{m}(x - x_k) \right)' + \prod_{\substack{k=1 \\ k \neq i}}^{m}(x - x_k) - \prod_{k=1}^{m}(x - x_k) \right), \quad \forall 1 \leq i \leq m.$$

因此，$(-1)^m L_{m+1}(x_i)$与$x_i \prod_{\substack{k=1 \\ k\neq i}}^{m}(x_i - x_k)$同号,从而

$$L_{m+1}(x_i)L_{m+1}(x_{i+1}) < 0, \text{于是，} \exists \xi_{i+1} \in (x_i, x_{i+1}),\ s.t. L_{m+1}(\xi_{i+1}) = 0,\ i = 1, 2, \cdots, m-1.$$

由$(-1)^m L_{m+1}(x_m)$与$\prod_{\substack{k=1 \\ k\neq i}}^{m}(x_m - x_k)$同号，而$L_{m+1}(x)$是首项系数为$(-1)^{m+1}$的$m+1$次多项式，可知

$$(-1)^m L_{m+1}(x_m) > 0,\ \lim_{x\to+\infty}(-1)^m L_{m+1}(x) = -\infty,$$

于是，$\exists \xi_{m+1} \in (x_m, +\infty), s.t. L_{m+1}(\xi_{m+1}) = 0.$

由$(-1)^m L_{m+1}(x_1)$与$\prod_{\substack{k=1 \\ k\neq i}}^{m}(x_1 - x_k)$同号，知$L_{m+1}(x_1) < 0$,而$L_{m+1}(0) = (m+1)! > 0$,于是

$$\exists \xi_1 \in (0, x_1), s.t. L_{m+1}(\xi_1) = 0.$$

综上，$L_{m+1}(x)$恰有$m+1$个正实根$(0 <)\xi_1 < \xi_2 < \cdots < \xi_{m+1}.$

4. 令$g(x) = \dfrac{x}{1+x^2} - f(x)$,由$0 \le f(x) \le \dfrac{x}{1+x^2}$,知

$$g(0) = \lim_{x\to+\infty} g(x) = 0, \quad \text{且} g(x) \ge 0, \forall x \ge 0.$$

若$g(x) \equiv 0$,则$g'(x) \equiv 0.$

若$g(x) \not\equiv 0$,则$\exists g(x_0) > 0.\ \exists M > x_0, s.t. \forall x > M$有$g(x) < g(x_0)$.于是

$\exists \xi \in (0, M), s.t. \quad g(\xi) = \max_{x\in[0.M]} g(x) > g(x_0) > g(0).$

由 Fermat 定理，$g'(\xi) = 0.$

5. $g'(x) \neq 0$,则$g(x) \neq g(y), \forall x \neq y$,且$g(+\infty) \neq g(-\infty).$

$$\text{令} F(x) = \begin{cases} f(\tan t) & |t| < \dfrac{\pi}{2} \\ f(\pm\infty) & t = \pm\dfrac{\pi}{2} \end{cases}; \quad G(x) = \begin{cases} g(\tan t) & |t| < \dfrac{\pi}{2} \\ g(\pm\infty) & t = \pm\dfrac{\pi}{2} \end{cases}.$$

对$\dfrac{F(\frac{\pi}{2}) - F(-\frac{\pi}{2})}{G(\frac{\pi}{2}) - G(-\frac{\pi}{2})}$使用*Cauchy*中值定理.

6. 令 $F(x)=\dfrac{f(x)}{g(x)}, G(x)=\dfrac{1}{g(x)}$, 对 $\dfrac{F(b)-F(a)}{G(b)-G(a)}$ 使用*Cauchy*中值定理.

7. $F(x)=\dfrac{f(x)}{x}, G(x)=\dfrac{1}{x}$, *Cauchy*中值定理.

8. $\dfrac{f(b)-f(a)}{b-a}=f'(\xi), \dfrac{f(b)-f(a)}{b^2-a^2}=\dfrac{f'(\eta)}{2\eta}$.

9. $f(0), f(1)$在点x处Taylor展开.

10. $xf'(\theta(x)x)=f(x)-f(0)=f'(0)x+\dfrac{f''(0)}{2}x^2+o(x^2)$,

$f'(\theta(x)x)-f'(0)=\dfrac{f''(0)}{2}x+o(x)$,

$\lim\limits_{x\to 0}\dfrac{f'(\theta(x)x)-f'(0)}{x}=\dfrac{f''(0)}{2}$.

$\lim\limits_{x\to 0}\theta(x)=\lim\limits_{x\to 0}\dfrac{\dfrac{f'(\theta(x)x)-f'(0)}{x}}{\dfrac{f'(\theta(x)x)-f'(0)}{\theta(x)x}}=\dfrac{\lim\limits_{x\to 0}\dfrac{f'(\theta(x)x)-f'(0)}{x}}{\lim\limits_{x\to 0}\dfrac{f'(\theta(x)x)-f'(0)}{\theta(x)x}}=\dfrac{\dfrac{f''(0)}{2}}{f''(0)}=\dfrac{1}{2}$.

11. 不妨设 $\exists x_0\in[1/2,1), s.t.\ -1=f(x_0)=\max\limits_{x\in[0,1]}f(x)$. 由 Cauchy 中值定理,

$$4\le\frac{f(1)-f(x_0)}{(1-x_0)^2}=\frac{f'(\xi_1)}{2(\xi_1-x_0)}=\frac{f'(\xi_1)-f'(x_0)}{2(\xi_1-x_0)}=\frac{f''(\xi)}{2}\quad(x_0<\xi_1<\xi<1).$$